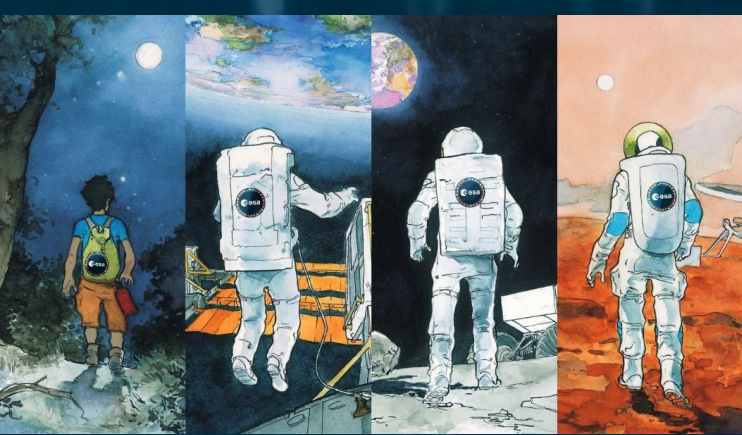


¹ ESA ESTEC, Noordwijk, The Netherlands (fabrice.cipriani@esa.int)

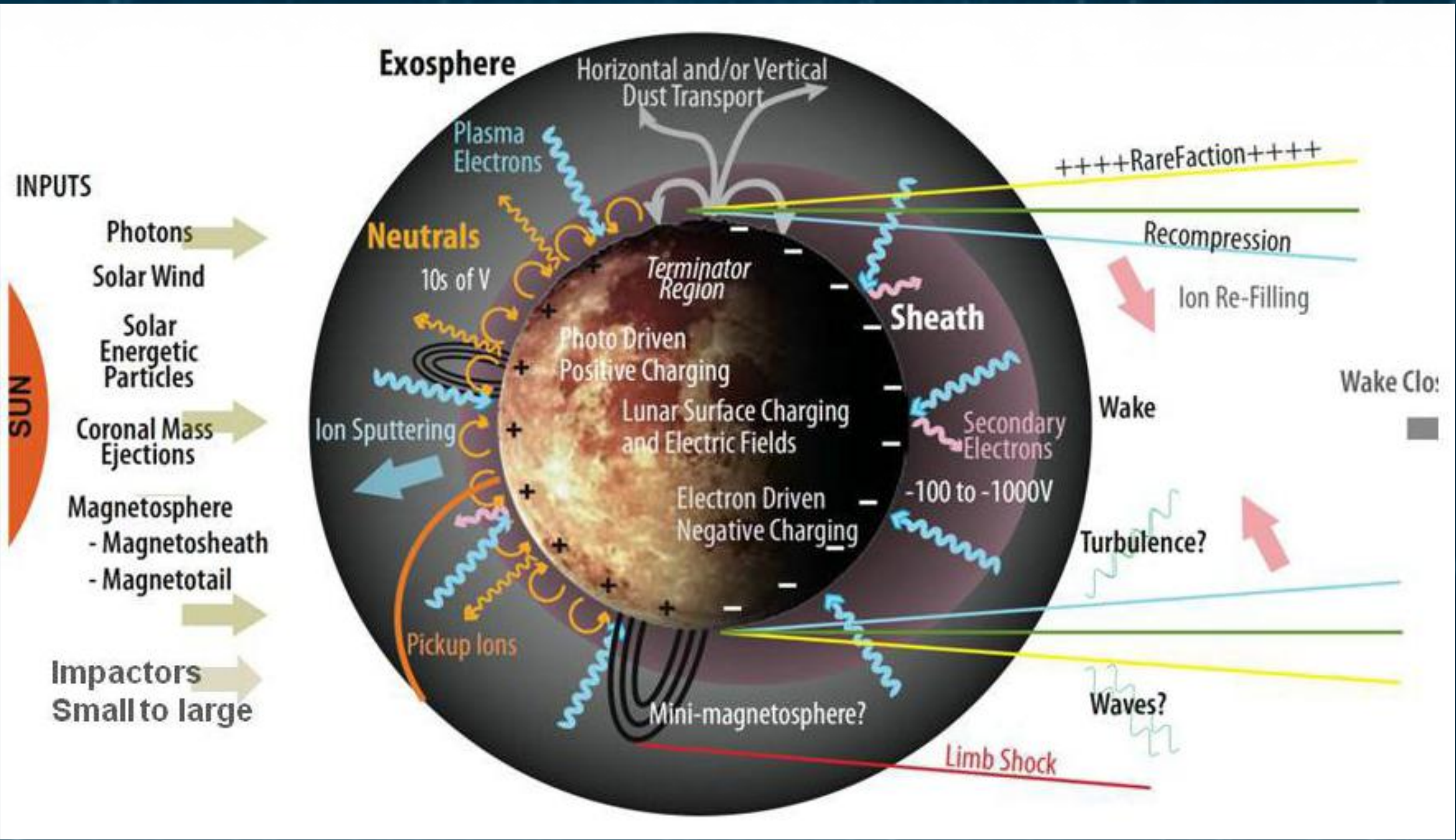
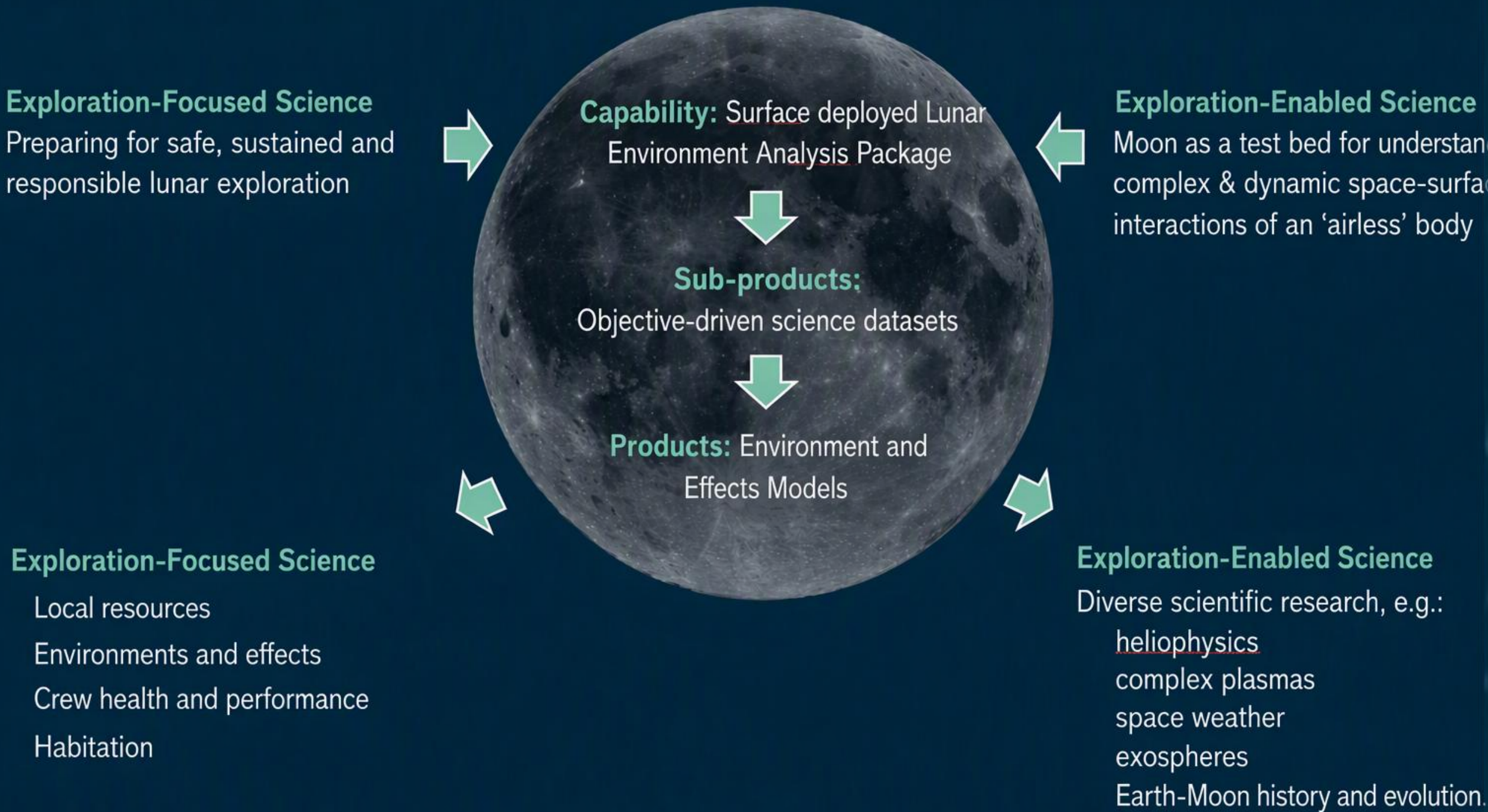
² AstroLEAP FDT Science Definition Report, 2024

(F. McDonald, Y. Futaana, I. Dandouras, S. Ranvier, V. Sterken, P. Fröhlich, H. Sargeant, R. Weber, F. Cipriani)

³ AstroLEAP Phase A/B1 Study team (IRF lead, FMI, IRS, BIRA, UKiel, UBern, CAS, IRAP)



A Lunar Surface Package to Monitor The Near Surface Lunar Environment



Exploration Science Context :

As part of ESA lunar exploration strategy[1], ESA is pursuing the development of enabling missions such as ARGONAUT[2] that will provide Europe with an access to the Lunar surface, and MoonLight constellation, establishing communication and navigation capabilities at the Lunar surface and between the Moon and Earth.

In this context, among Lunar Exploration Focused Science themes[4], the in-situ characterization of the near surface lunar environments and their interactions with landed assets becomes one of the priorities with the goal to de-risk lunar surface hazards and facilitate the design of future human and robotic surface missions, while enabling timely opportunities to perform cross-disciplinary surface science, of interest to the Heliophysics community.

The ECHO (previously AstroLEAP) phase A/B1 study, is targeting the definition of an integrated in-situ environment observatory for the lunar surface aiming at characterizing primarily plasma populations, the lunar exosphere, transported lunar dust and regolith properties, the incoming meteoritic flux and secondary ejecta populations, as well as the electro-magnetic environment and radiation is currently running until end 2026, with a follow-up implementation phase B2/C/D planned to be Kicked-Off during 2027.

A tentative Lunar Surface deployment is foreseen with one of the Argonaut missions to occur in the period 2031-2033.

Science Opportunities :

A Science team involvement in support of the next phase will be possible through an **open call** to be released early 2027. This call will target the establishment of a **Science Team** providing an advisory support to ESA and the development team (including instruments scientists) for exploration focused science related aspects including the fine-tuning of scientific priorities, operational scenarios consolidation, data products definition and science performance assessment throughout Phase B2/C/D and E.

Questions of interest to and synergies with the Heliophysics community that can be supported by lunar surface measurements will be highly relevant and welcome (please just reach out if you want to interact / know more !!).

High-level Investigation science Themes addressed by the observatory (ongoing Phase A/B1, IRF led consortium)

- Near-surface plasma particles and fields
- Influence from Earth's ionospheric/magnetospheric plasma
- Energetic particles environment
- Surface-bounded exosphere
- Dust Environment
- Human exploration impact

Analytical Technics and Heritage instruments : includes currently Plasma Analyzer (Solar Wind and magnetospheric ions and electrons), (fast and slow moving) dust instrument, DC electric-field and wave instrument, neutral mass spectrometer, radiation monitor, charged regolith sensor

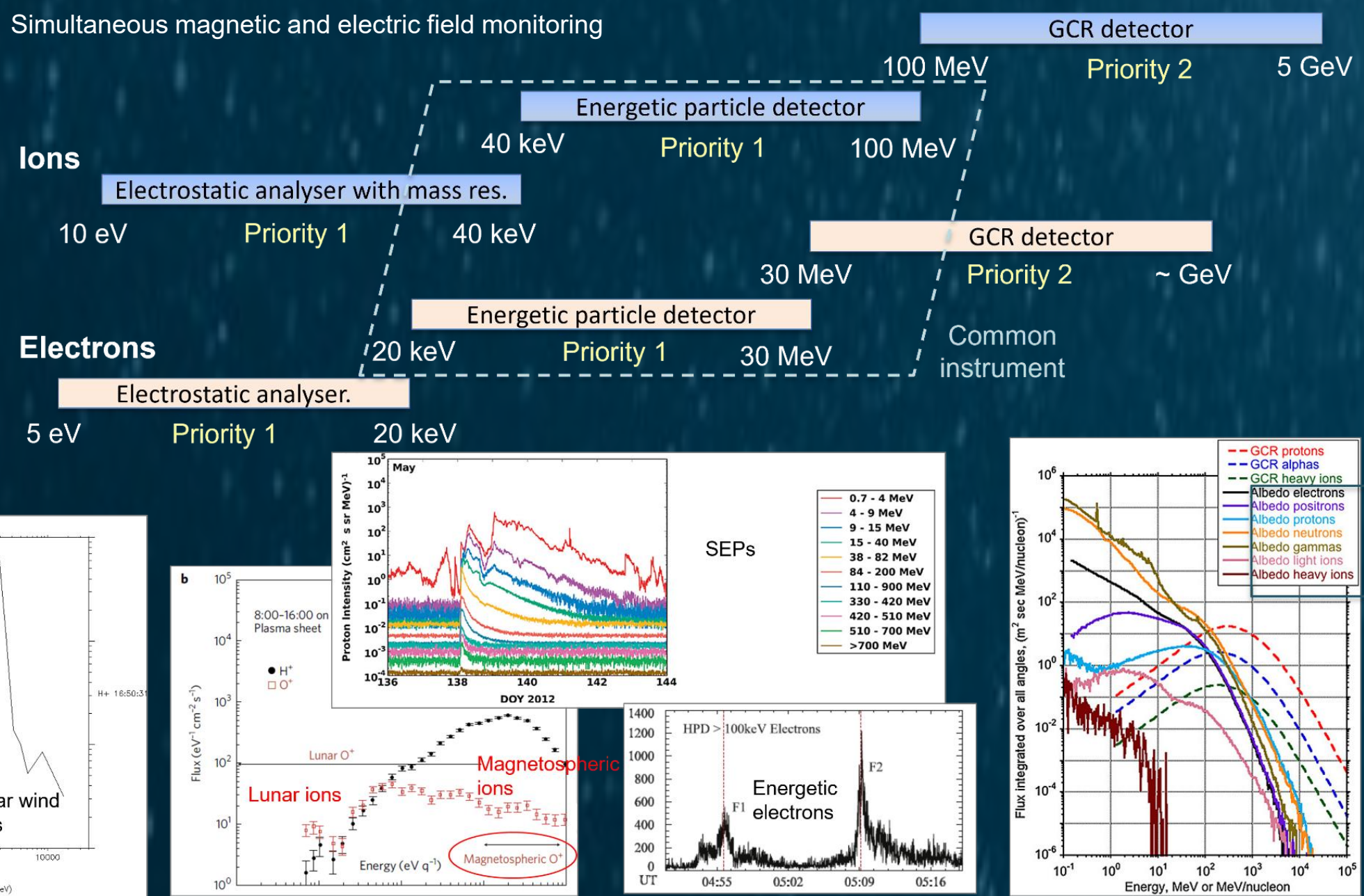
- **Facility Definition Team Science Report (end 2024)**
- **Topical Discussion Meeting held at European Space Weather Week in Nov. 2025**
- **Lunar Environment Science Workshop held February 2026**
- **ECHO (previously) Science Team call early 2027**
- **2d Science Workshop in 2028 (TBC)**

Theme 1: Space plasma, energetic particles, lunar exosphere

Plasma particles and fields: Complex properties and behaviours; a driver for surface sputtering and charging, and exosphere dynamics (with solar UV)

Solar wind and Earth magnetospheric environments: Effects and large variability (e.g. Earth's magnetotail), surface sputtering and charging.

Energetic particles (pristine / albedo): Radiation environment; surface hydration sensitivity.

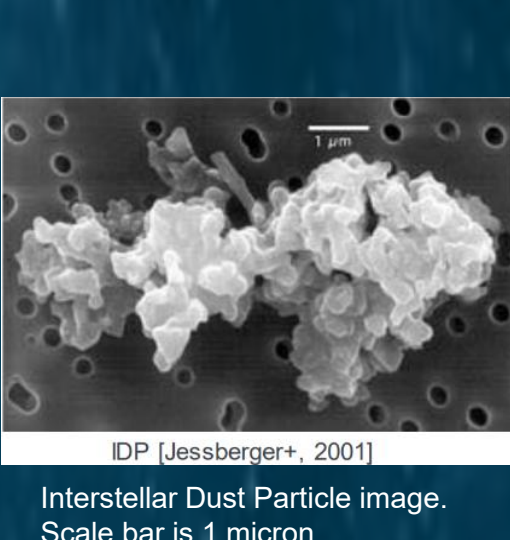
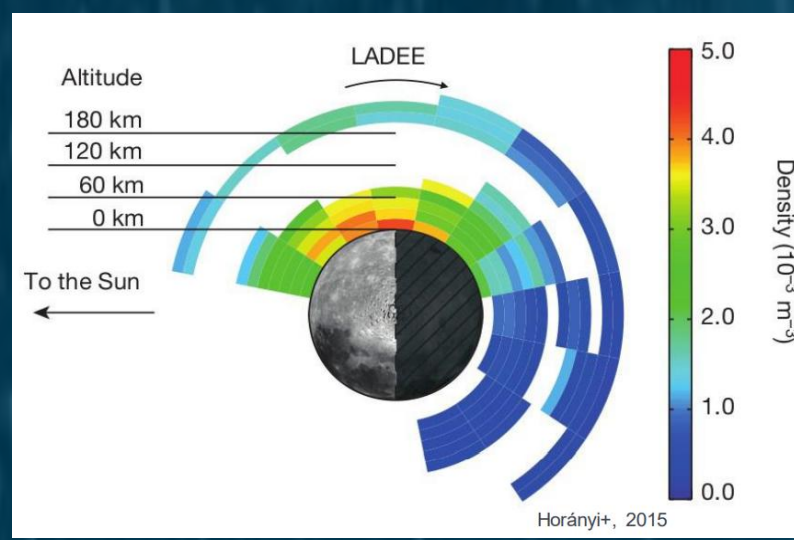
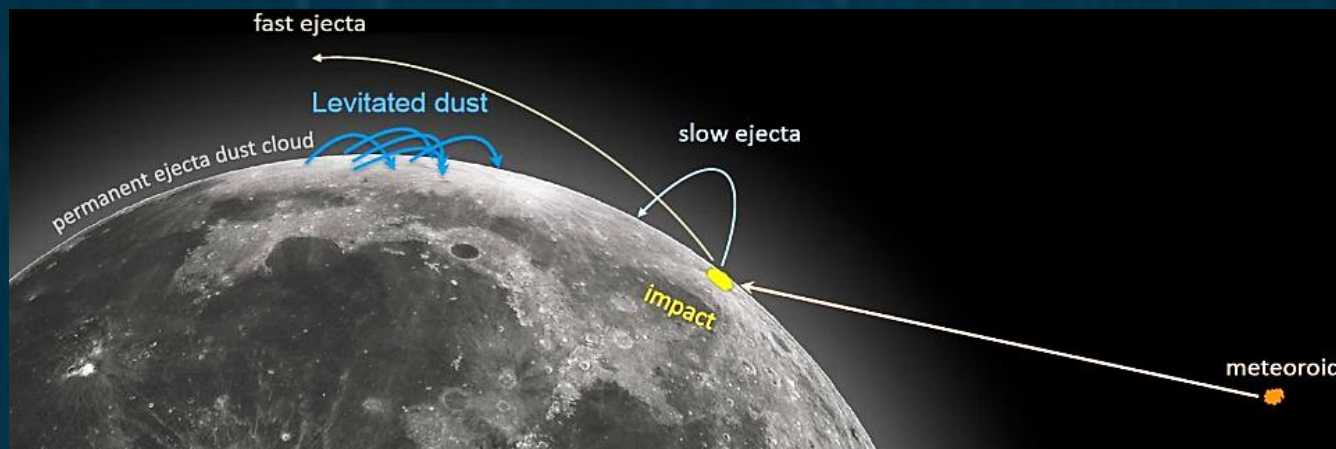


Theme 2 : local and external Dust populations, coupling with plasma and fields

Dust Environment: IDPs, impact ejecta, locally lofted/levitated dust fluxes, behaviours over temporally variable conditions.

Surface-bounded exosphere: Contribution from dust impacts to the exosphere; Lunar dust exosphere.

Plasma particles and fields: plasma-regolith coupling; electrostatic transport



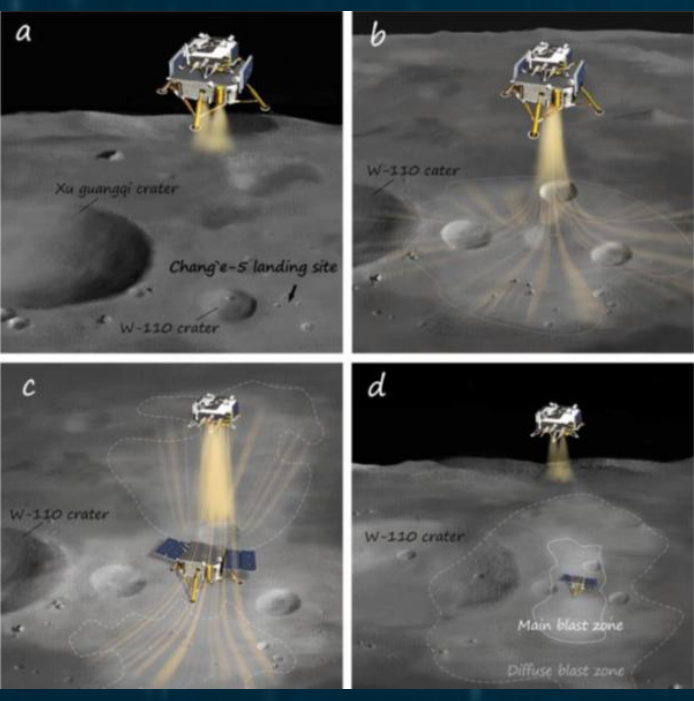
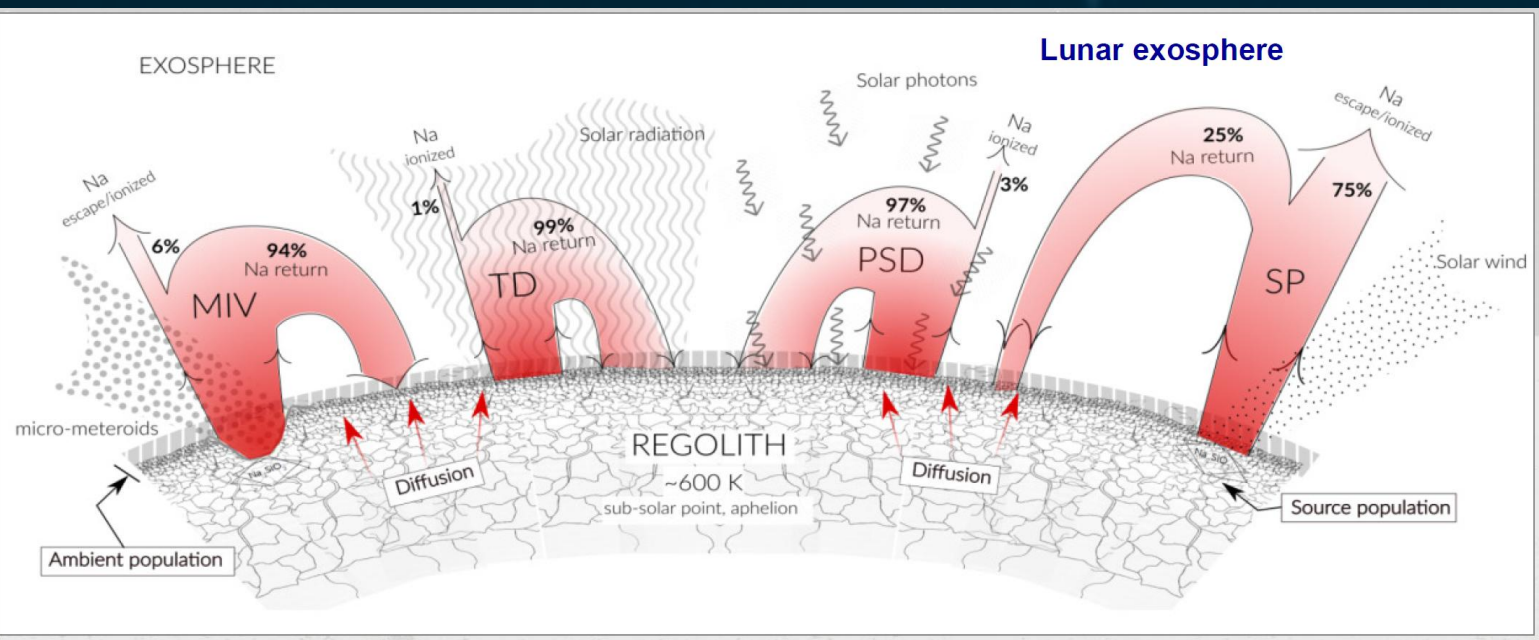
	Regolith	Naturally lofted	Anthrop. lofted	Ejecta	IDP/ISD
Flux / count	-	-	-	-	-
Charge	-	-	-	-	-
Speed	-	-	-	-	-
a/m	-	-	-	-	-
Direction	-	-	-	-	-
Height profile	-	-	-	-	-
Mass	-	m ⁻² s ⁻¹ (g/m)	-	-	-
Size	-	r ⁻² m ² /hr	-	-	-
Surface Potential	-	Phi ⁻¹ gr	-	-	-
Density	-	-	-	-	-
Cohesion/adhesion	-	-	-	-	-
EL conductivity	-	-	-	-	-
	Primary	Secondary	Calculate from other measurements		

Theme 3: Exosphere, Volatiles, Human Exploration impact

Human Exploration Impact: Transfer of life and other contaminants (e.g. volatiles, organics...); outgassing; disturbance/contamination and redistribution.

Dust Environment: Dust mitigation preparation for exploration; monitoring the impact from human activity, e.g. regolith contamination, triboelectric charging...

Surface-bounded exosphere: Exosphere characterisation before further perturbed by exploration; Onward monitoring of human exploration impact.



Exosphere: density, composition... e.g. neutral mass spectrometer

Dust contamination: dust deposition density, e.g. solar cells, dust microbalances (+/- optical sensors)...

Theme 4: Surface potentials and EM fields

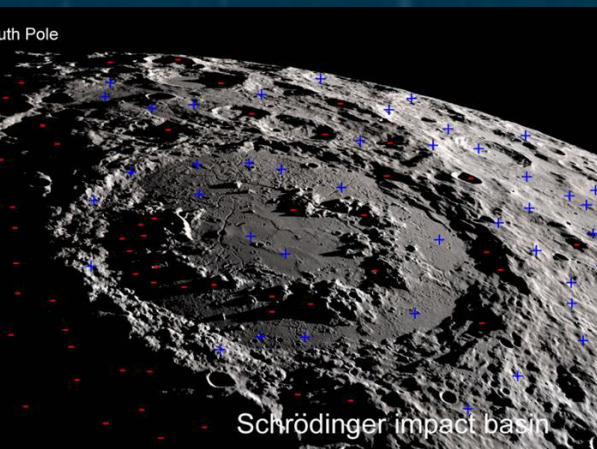
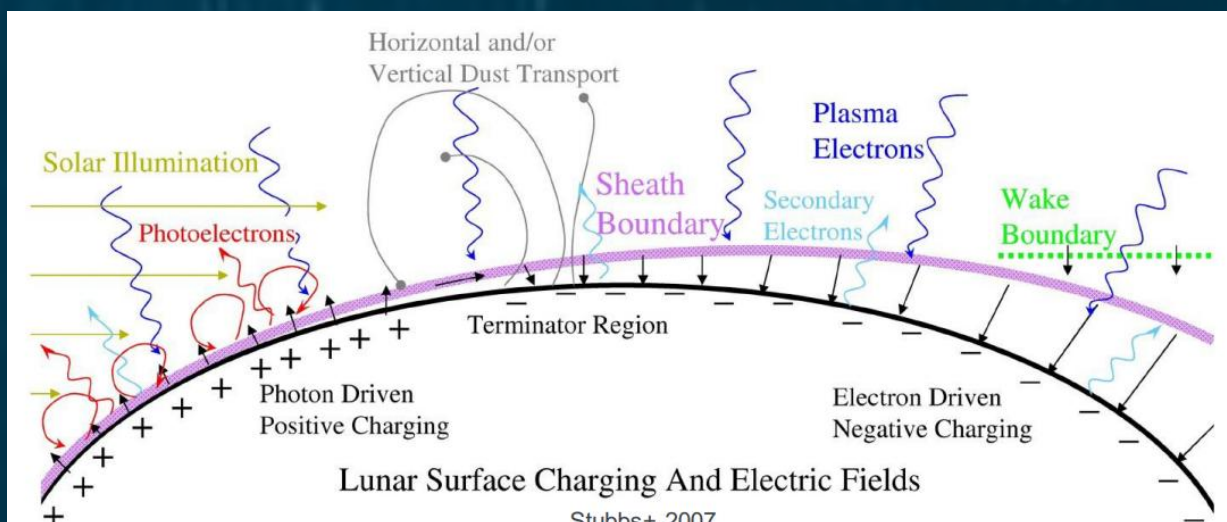
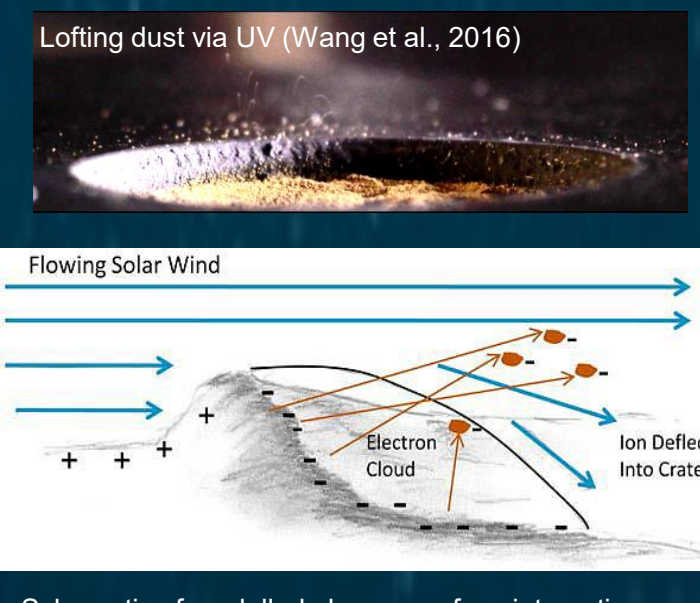
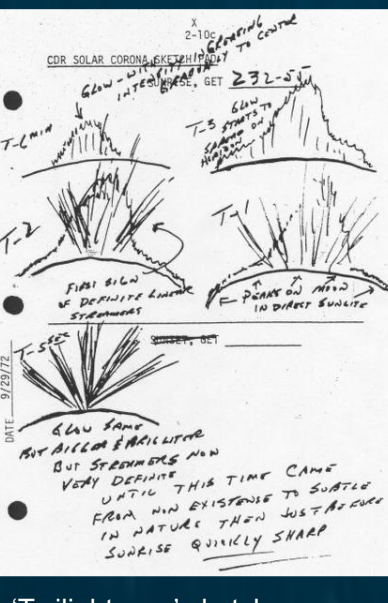
Dust Environment: Investigate the purported lofted / levitating dust and associated mechanisms; inform dust mitigation.

Plasma particles and fields:

Complex plasma-surface coupling: 'dusty plasma'.

Earth magnetospheric environments: Effects and modification when passing through the Earth's magnetotail.

Human exploration impact: hazardous charging / discharging of surface assets (incl. astronauts suits), interactions with dust (performance degradation).



Efield monitoring:
DC: 0.1 - ~x100V/m
AC: 0.02 -1KHz (dust dynamics)
AC: 0.02 -400KHz (dust & plasma)
Sensors distributed horizontally and vertically.